RemoVia

Setup & TV Pairing Guide - beginner edition

RemoVia is one remote on your phone for every smart TV in the house - Roku, Samsung, Google TV, Android TV and Fire TV. This guide is written for beginners: it tells you exactly what to press, what you will see when it works, and what to do if it doesn't. Nothing here can break your TV or your computer. The same guide is on the web at iecincglobal.com/removia-guide.html.

Before you start - five words we will use

Bridge: a small program that runs on your Mac and talks to your TVs for you. Your phone tells the bridge; the bridge tells the TV.

Terminal: a window on the Mac where you type commands. Don't worry - we tell you exactly what to type, word for word.

Token: a long secret password that your bridge makes for you. It proves the phone asking is yours.

IP address: your TV's number on your home Wi-Fi - like a house number on a street. We show you where the TV displays its own number.

Tailscale: a free app that builds a private, safe connection between your phone and your Mac, so the remote works from anywhere.

Part 1 - What you need

- A Mac on the same Wi-Fi as your TVs. The bridge runs on it. (The Mac needs Python 3.11 or newer - the bridge sets up everything else by itself the first time it runs.)
- Your phone. The remote is one web page - nothing to install from an app store for the remote itself.
- The RemoVia bridge ZIP file (the download). Unzipping it makes a folder called removia-main.
- A free Tailscale account. You create it by signing in during Part 4 - no separate sign-up trip.
- Each TV's own remote control. Pairing is approved on the TV screen, once per TV.

TVs RemoVia works with: Roku - Samsung - Google TV and Android TV sets (Sony, TCL, Hisense, Philips, Xiaomi, NVIDIA Shield) - Amazon Fire TV. LG (webOS) sets are found and saved, but cannot be controlled yet.

CHECKPOINT: You should now have: the unzipped removia-main folder in your Downloads, your phone in hand, and each TV's own remote nearby.

Part 2 - Open Terminal (the only 'computer part')

Terminal sounds technical. It is just a window where you type. You will type a handful of short lines in this whole guide, and we give you every one of them exactly.

1. On the Mac, click the magnifying glass at the top-right of the screen (that's called Spotlight). Type

the word Terminal. Press Enter.

YOU SHOULD SEE: A plain window opens with a few lines of text and a blinking cursor. That window is Terminal. You did it.

2. Know this: you can copy and paste every command in this guide instead of typing - copy with Cmd+C, paste into Terminal with Cmd+V, then press Enter to run it.
3. Know this too: nothing you type here can break your TV or your computer. The worst that can happen is an error message - nothing changes, and you just try again.

CHECKPOINT: You should now see: a Terminal window open on your Mac, waiting for you to type.

Part 3 - Start the bridge, and meet your token

1. Type this exactly, then press Enter:

```
cd ~/Downloads/removia-main
```

YOU SHOULD SEE: The text at the left of the window now ends with removia-main. That means you are inside the folder.

IF NOT: If it says 'no such file or folder', the ZIP was not unzipped yet. Go to Downloads, double-click the ZIP, then try the line again.

2. Type this exactly, then press Enter:

```
./start.sh
```

YOU SHOULD SEE: The first time, several minutes of text scroll by while the bridge builds itself - that is normal. It is finished when a line says 'Uvicorn running on http://0.0.0.0:8765'. That line means: the bridge is ON.

IF NOT: The Mac may pop up a question - 'allow Python to accept incoming connections?' Click Allow. If the text ends with 'address already in use', an old copy of the bridge is already running - see Troubleshooting in Part 7.

3. Your token appears once. In that first run of text, look for the words:

```
First run: your RemoVia bridge access token is:
```

Under those words is a long line of letters and numbers. That is your token. Your bridge made it, on your computer, for you - nobody else has it, and IEC cannot see it. Select that long line, copy it (Cmd+C), and paste it somewhere safe, like your Notes app. You will paste it into the remote page in Part 4.

If you lose it: the token also lives in a settings file on your Mac called config.json, inside a folder named .tv-remote in your home folder. Open that file only to read your token. Never send that file, or the token, to anyone - not even to us. Whoever has the token and a path to your bridge can control your TVs.

4. Leave this Terminal window open. The bridge runs while the window stays open. Closing the window turns the remote off - that is normal, not broken. To turn it back on, open Terminal and run the same two lines from steps 1 and 2 again.

CHECKPOINT: You should now see: 'Uvicorn running' in Terminal, the window still open, and your token saved somewhere safe.

Part 4 - Connect your phone securely (Tailscale)

Why this part exists, in one sentence: the remote page (removia.iecincglobal.com) is a secure web page, and browsers will not let a secure page talk directly to the bridge in your house. Tailscale is the private path between them - think of it as a locked corridor that only your devices can enter. It is free.

4A - On your phone

1. Open the App Store. Search for Tailscale. Tap Get, then Open.
2. Sign in. Use the same account you will use on the Mac (for example, the same Google account). The account is created the first time you sign in.

   YOU SHOULD SEE: Your phone warns you that Tailscale wants to add a VPN configuration. This is expected - the private corridor is what the warning is about. Tap Allow, and type your phone passcode if it asks.

   YOU SHOULD SEE: The Tailscale app says Connected and lists your phone.

4B - On the Mac

1. Install Tailscale on the Mac as well (from tailscale.com/download or the Mac App Store), open it, and sign in with the same account as the phone.

   YOU SHOULD SEE: A short welcome survey may appear ('Help us better understand your product needs'). For 'What is your primary reason for using Tailscale?' pick Personal or At-Home Use, pick any answer for your role, and press Submit. It changes nothing about your setup.

   YOU SHOULD SEE: The Mac app says Connected. Your Mac and your phone now both appear in the device list.

4C - Turn on HTTPS (one time only)

1. In your web browser on the Mac, go to console.tailscale.com/admin/dns. Sign in with the same account if it asks.
2. Scroll to HTTPS Certificates and click Enable HTTPS.

   YOU SHOULD SEE: The section now shows HTTPS as enabled. You never need this page again.

4D - Give the bridge its secure address

1. Open a NEW Terminal window (Terminal's Window menu, or Cmd+N) - leave the bridge window from Part 3 untouched. Type this exactly, then press Enter:

```
tailscale serve --bg 8765
```

   YOU SHOULD SEE: The words 'Serve started and running in the background', and an address that looks like https://your-mac.tail1234.ts.net. Write this address down - it is your bridge's private web address.

   IF NOT: Hiccup 1: the command may stop and say 'Serve is not enabled on your tailnet. To enable, visit:' followed by a link. That is the one-time switch from Part 4C asking to be

turned on. Hold Cmd and click the link in Terminal, sign in, click Enable - then type the same command again and press Enter.

IF NOT: Hiccup 2: a message saying 'Another client is changing the serve config' or 'etag mismatch - please try again' simply means two changes bumped into each other. Run the same command once more. It works.

4E - Tell the remote page where your bridge lives

1. On your phone, open removia.iecincglobal.com. Tap the gear (Settings) at the top-right.
2. In Bridge address, paste the ts.net address from 4D. In Token, paste your bridge token from Part 3. Tap Save. Then tap Test.

   YOU SHOULD SEE: The dot at the top of the page turns green and the word Connected appears. Your remote now works from anywhere - home or away - and nothing in your house is exposed to the public internet.

CHECKPOINT: You should now see: a green dot and 'Connected' on the RemoVia page on your phone, and both your phone and your Mac listed as Connected in Tailscale.

Part 5 - Add your TV, and find its number (IP address)

1. In RemoVia on your phone, tap + TV, then tap Scan the network.

   YOU SHOULD SEE: After a few seconds, your TV appears in a list. Tap Add next to it, type a room name (for example BedRoom), and save.

   IF NOT: If the scan finds nothing, the TV may be asleep or on a different Wi-Fi. Wake it, check it is on the same Wi-Fi as the Mac, and scan again. Still nothing? Add it by hand - the next steps show how.
2. To add by hand, you need the TV's IP address - its house number on your Wi-Fi. The TV displays it in its own settings. Samsung: with the TV's own remote, press Settings > General > Network > Network Status > IP Settings. You will see a screen listing IP Address, Subnet Mask and Gateway. Write down the IP Address line. Then press Cancel to leave without changing anything. Roku: with the TV's own remote, press Settings > Network > About. Write down the IP address shown there.
3. Back in RemoVia's + TV screen, pick the TV's brand, type the IP address you wrote down, type a room name, and tap Add this TV.

The one mistake almost everyone makes.
Entering the Gateway number instead of the IP Address. The Gateway is your router's number - on most home networks it is 10.0.0.1. The two lines sit next to each other on the TV screen. If you enter the gateway, the bridge knocks on the router's door and pairing fails with ConnectionRefusedError. Always use the line labelled IP Address.

CHECKPOINT: You should now see: your TV as a named button (for example 'BedRoom') at the top of the RemoVia page.

Part 6 - Pair by brand

Pairing means the TV agrees to listen to RemoVia. It happens once per TV - after that, the TV

remembers and you never do this again.

Roku - no pairing, one setting on the TV

There is nothing to pair in the app. But the Roku must be allowed to listen. This is done on the TV, with the TV's own remote - not in RemoVia: Settings > System > Advanced system settings > Control by mobile apps > Network access, then choose Permissive.

YOU SHOULD SEE: Text under it saying control by mobile apps 'is enabled always, but accepts commands only from private network addresses or the same subnet'. That sentence is the good one - it means apps on your home network may control the TV, and nothing outside can.

IF NOT: If Network access is left on Limited, every button in RemoVia answers with error (403).

CHECKPOINT: Test it: in RemoVia, select the Roku and press Home. The TV should jump to its home screen.

Samsung - press Allow on the TV

Read this first.
The pairing prompt appears on the TV, not on your phone - and it only stays on screen a short time. Have the TV's own remote in your hand BEFORE you tap Start pairing. The TV must be fully on, showing a normal channel or app, with all menus closed.

1. On the phone: in RemoVia, select your Samsung TV, then tap Start pairing.

   YOU SHOULD SEE: On the TV: a box pops up saying 'RemoVia - Your permission is required to connect your device to TV.' The top choice is Allow.
2. On the TV remote: press the middle / OK button to choose Allow.
3. On the phone: tap Done.
4. Test it: in RemoVia, press VOL +. The volume bar should move on the TV. Pairing is done - forever.

   IF NOT: If the phone says the TV 'did not accept in time', the prompt timed out or a menu was open. Close all menus on the TV and repeat from step 1, remote in hand. If no prompt ever appears: on the TV, open Settings > General > External Device Manager > Device Connect Manager and set Access Notification to Always On, then try again. (Also under Settings > General > Network > Expert Settings, 'Power with Mobile' should be on. Menu names vary a little by model.)

CHECKPOINT: You should now see: the Samsung's volume move when you press VOL + in RemoVia.

Google TV / Android TV - type the PIN the TV shows you

For Sony, TCL, Hisense, Philips, Xiaomi and NVIDIA Shield sets:

1. On the phone: select the TV and tap Start pairing.

   YOU SHOULD SEE: On the TV: a 6-digit PIN - six numbers on the TV screen.
2. On the phone: type those six numbers into RemoVia and tap Pair with this PIN.

CHECKPOINT: Test it: press any button in RemoVia - the TV responds.

Fire TV - one TV setting, then Allow

1. On the TV remote: Settings > My Fire TV > Developer Options > ADB Debugging > On. (This simply lets the TV accept a remote connection.)
2. On the phone: select the TV and tap Start pairing.

   YOU SHOULD SEE: On the TV: a prompt asking to allow the connection. Tick 'Always allow' if it is shown, then choose Allow.
3. On the phone: tap Done.

CHECKPOINT: Test it: press any button in RemoVia - the TV responds.

LG (webOS)

RemoVia can find an LG TV and save it, but it cannot control it yet - there is no pairing step. If you have an LG, skip it for now; the other brands above are fully supported.

Part 7 - Troubleshooting

Find the words you see in bold. Every fix is short, and none of them can harm anything.

Error (403) on a Roku
Fix: On the TV: Settings > System > Advanced system settings > Control by mobile apps > Network access = Permissive. (Part 6, Roku.)

ConnectionRefusedError
Fix: The bridge used the wrong IP (often the gateway, 10.0.0.1), or the TV is off. Read the TV's real IP Address in its Network settings (Part 5), re-add the TV with that address, and make sure the TV is on.

"The TV did not accept in time, or it is off"
Fix: TV fully on, all menus closed, TV remote in hand. Tap Start pairing and press Allow on the TV promptly.

"Stopped answering"
Fix: Restart the bridge: click its Terminal window, press Ctrl+C, then type ./start.sh and press Enter.

"Address already in use" when starting the bridge
Fix: The old bridge is still running. Stop it first: Ctrl+C in its Terminal window. Or, in a new Terminal window, type: kill $(lsof -ti :8765) and press Enter. Then run ./start.sh again.

"No such file or folder" after the cd line
Fix: The bridge folder is not where Terminal is looking - usually the ZIP was never unzipped. In Downloads, double-click the ZIP so the removia-main folder appears. Then run the cd line again.

Red dot - the page cannot reach the bridge
Fix: Check Tailscale says Connected on both phone and Mac, with the same account. Check the bridge Terminal window is open and says 'Uvicorn running'. Check the Bridge address and Token in the page's Settings match Part 4 exactly.

Phone cannot connect at home

Fix: The Mac's firewall may be blocking the bridge. On the Mac: System Settings > Network > Firewall > Options > allow incoming connections for Python.

Still stuck? Go back to the last CHECKPOINT you passed and re-read that part's YOU SHOULD SEE lines. Almost every problem is one of the rows above.

Part 8 - Safety notes

- Your token is a key to your TVs. Your bridge made it, on your computer. Never post it, share it, or share the ~/.tv-remote/config.json file that holds it - with anyone, including us.
- Tailscale keeps everything private. Only devices signed in to your account can reach your bridge. Nothing in your house is exposed to the public internet, and no router port-forwarding is needed - never set up port-forwarding for the bridge.
- Pairing prompts appear on the TV itself. Press Allow only for a prompt you just started. The prompt names the app - RemoVia.
- The bridge window is the on/off switch. Bridge window open = remote on. Window closed = remote off. That is normal, and reopening it (Part 3, steps 1-2) turns everything back on - your TVs stay paired.

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